5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · AI IN MEDIA SERVICES

AI in Media Services

Evaluating AI and ML for media, and the traffic of AI media services

3GPP

OCTOBER 2026 – www.5g-mag.com

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About the project

The documents the project rests on

3GPP TR 26.927

Technical Report

Study on AI and ML in 5G media services: media-based applications and the media service architecture for AI/ML.

3GPP TR 26.847

Technical Report

Evaluation of AI and ML in 5G media services: feasibility evaluations for a given set of services.

3GPP TR 26.870

Technical Report, draft

Study on Media Aspects for 6G System, which records the traffic of AI media services measured with the testbed.

FIND OUT MORE **Technical Analysis** **Specifications**

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What the reports cover

The SA4 reports on AI and ML in media, and the 6G media study that the traffic measurements feed

TR 26.927 media-based applications and the architecture for AI/ML

- Object recognition in image and video
- Video quality enhancement in streaming
- Crowd-sourced media capture
- Natural language processing on speech
- Media service architecture for AI/ML
- Split inference
- Model delivery

TR 26.847 the evaluation of the services

- Testbed architectures and anchors
- Split inferencing
- Evaluation metrics
- Compressed model transfer, speech recognition
- Object detection and labelling
- Bit-incremental model transmission
- Real-time translation over IMS
- Model-data transmission

TR 26.870, draft the seven work topics of the 6G media study; only AI traffic characteristics have findings so far

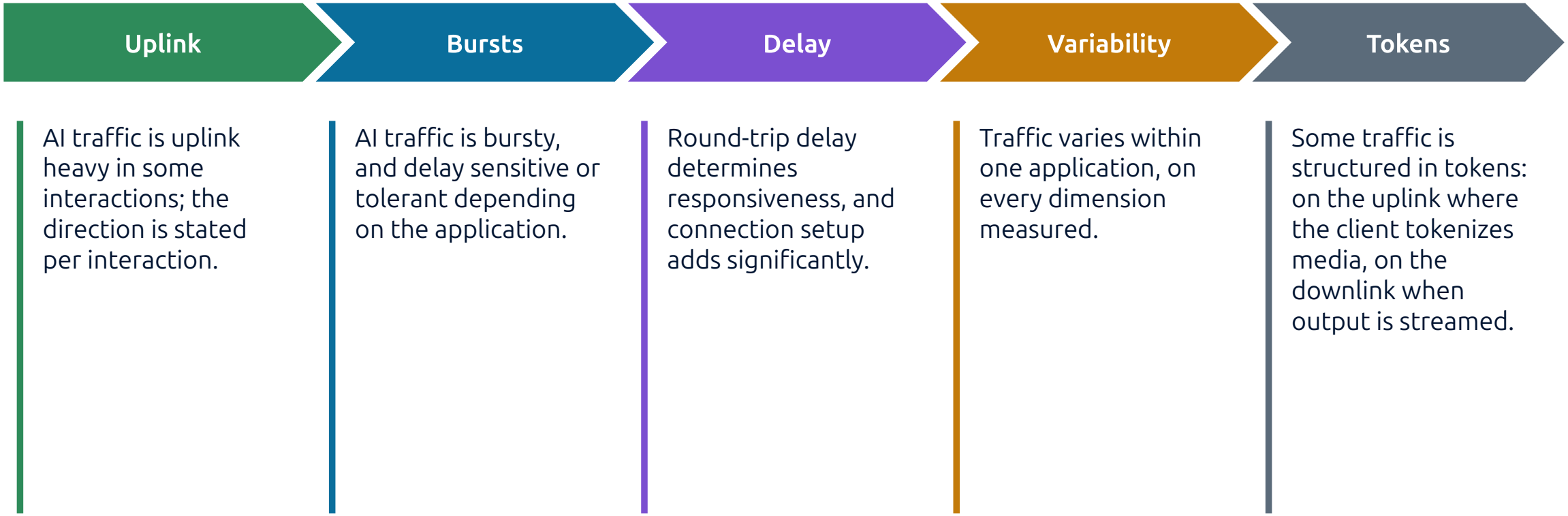
- Media Delivery Architecture
- 6G media
- Media aspects related to SA2 topics
- Media for ubiquitous access
- Trusted and private communication for media
- Software and asset management
- Real-time communication system enhancements

AI/ML in media is studied in SA4; the network-side analytics of AI/ML (NWDAF) are studied in SA2.

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AI traffic characteristics

The five initial findings of the 6G media study, measured with the 6G AI Traffic Characterization testbed



The findings are initial, from a draft report, and are to be updated as more measurement results become available.

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The testbed

How the 6G AI Traffic Characterization testbed runs an interaction and measures its traffic

Orchestrator

The Orchestrator drives the interactions (chat, agents, image, video, audio) and the MCP tool servers.

LLM client layer

Carries the requests to the LLM and tool APIs, with L7 capture by mitmproxy.

Network emulator

Applies the tc/netem profiles to the traffic before it reaches the Internet.

Capture and analysis

L3/L4 and L7 capture feed the TrafficLogger and the MetricsCalculator.

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What it measures

Eleven interaction patterns, run over emulated network profiles, reported in four metric categories

Interaction patterns from chat to agent-to-agent collaboration

- Chat
- Agent with MCP
- Image generation
- Multimodal analysis
- Multimodal analysis, live AI mode
- Video understanding
- Real-time text and audio
- Direct web search
- Computer use
- Personal assistant agent
- Agent-to-agent collaboration

Network profiles deployment conditions, plus a lossy profile that finds where an interaction breaks

- ideal
- 6g_itu_hrllc
- 5g_urban
- wifi_good
- cell_edge
- satellite_leo
- satellite_geo
- congested
- 5qi_7
- 5qi_80
- lossy

Metrics recorded per request and from packet captures

- A. Traffic characteristics
- B. Response time and errors
- C. Service metrics
- D. Task-related QoE

A test run takes one interaction pattern, one network profile and one execution mode, and produces a metric record.

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How it grew

What each 3GPP release added to AI and ML in media services

Release 18

The SA4 study FS_AI4Media is listed in TR 21.918 without a description. The SA1 work item AIML_MT covers the traffic of AI/ML model transfer.

Release 19

TR 26.927 and TR 26.847 are approved at SA#108 and created as version 19.0.0.

Release 20

Work plan of 28 September 2026: the reports recommend stage 3 work on AI/ML in IMS and a new evaluation study. The 6G media study, TR 26.870, is a draft (V0.6.1). Scopes unverified.

From the Standards pages of the website; the Release 20 items come from the 3GPP work plan, with no Release 20 release description (TR 21.920) available.

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Repositories

The repositories of the AI traffic and AI/ML evaluation work, from their READMEs

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
6G-Testbed	the 6G AI Traffic Characterization testbed: measures AI service traffic under network conditions	AI traffic testbed Network emulation	none yet	GitHub
netemu-dashboard	a web UI and REST API for network impairment profiles	Network emulator control	none yet	GitHub
rt-ai-ml-evaluation-framework	the AI and ML evaluation framework of the 3GPP SA4 report on AI and ML in 5G media services	Compressed model transfer Split inferencing Translation scenario	none yet	GitHub

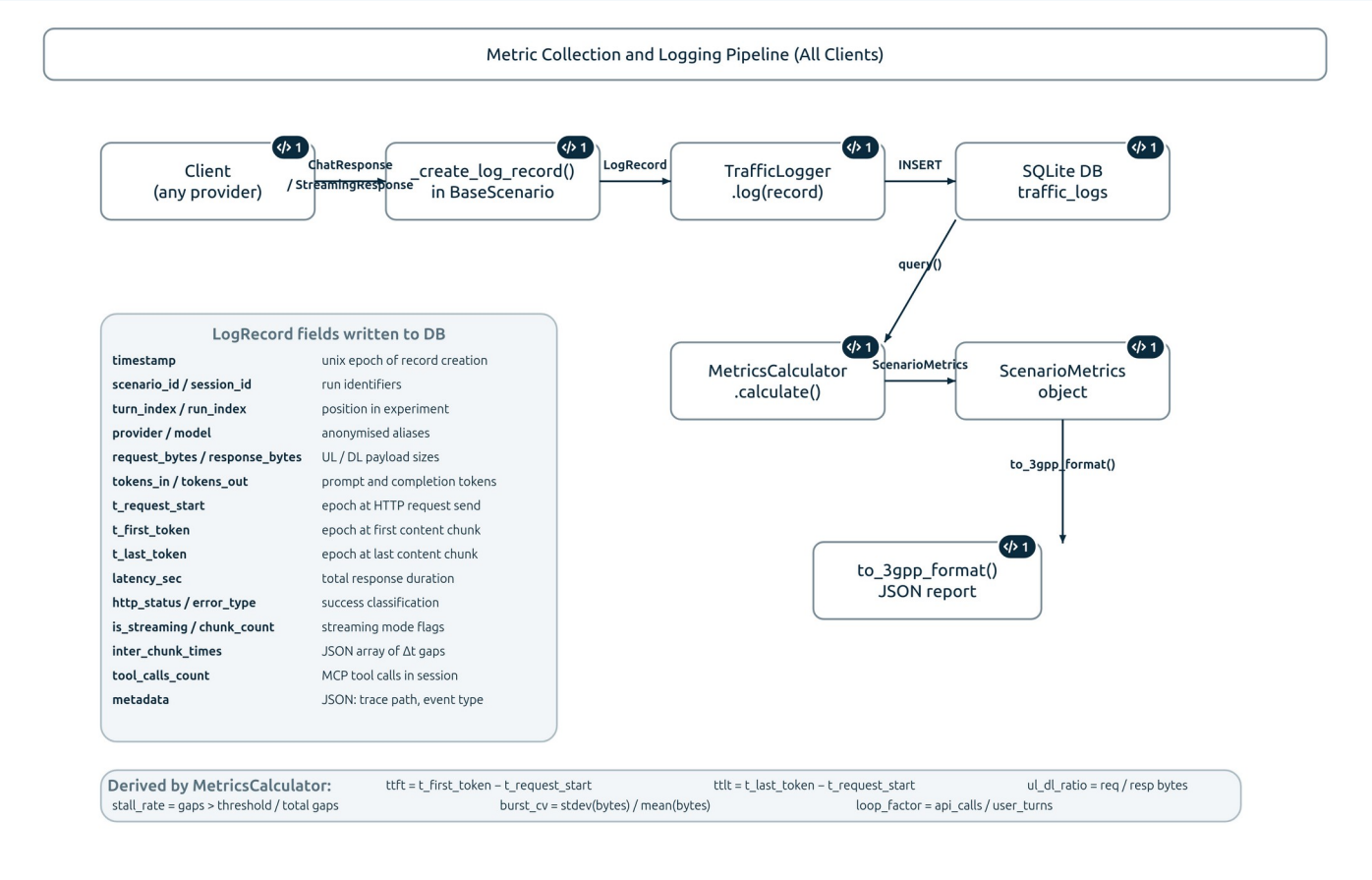
FIND OUT MORE Technical Analysis Specifications

● released ○ no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 6G-Testbed

All seven pipeline boxes are in the 6G-Testbed repository (aitestbed).

CONTRIBUTE

Latest updates and roadmap

Status on 22 October 2026

IN DEVELOPMENT

AI traffic testbed

Agentic measurements for the 6G media study, and video analysis and chat traffic with tokens.

Network emulator

Network capture and measurement as a standalone package, which the testbed uses for shaping and pcap analysis.

AI and ML evaluation framework

Split inferencing and compressed model transfer, including incremental transfer of model data.

UNDER STUDY

Metrics

A careful review and detailed definition of the metrics, and of how the traffic characteristics are measured.

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GitHub

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